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\documentclass[11pt]{article}
\usepackage[T1]{fontenc}
\usepackage[utf8]{inputenc}
\usepackage{lmodern}
\usepackage{amsmath,amssymb,booktabs}
\usepackage[a4paper,margin=1in]{geometry}

\title{Unified Quartic--Sextic Axis-Representation Transform Program}
\author{}
\date{}

\begin{document}
\maketitle

\section*{Scope}
A single Python GUI program was implemented to replace the previous quartic-only Yamada app. The new
\begin{itemize}
\item quartic Yamada transform in A reduction,
\item quartic Yamada-like transform in S reduction,
\item quartic tensor algorithm (Watson  $\rightarrow \tau \rightarrow$  permutation  $\rightarrow$  Watson)
\item sextic tensor algorithm (Watson  $\rightarrow \phi \rightarrow$  permutation  $\rightarrow$  Watson)
\end{itemize}

Representations are  $I, II, III$ . Reduction is preserved (input reduction equals output reduction).

\section*{Implemented Quartic Models}
For quartic constants
\[\mathbf{D}_A=(A_J,A_{JK},A_K,d_J,d_K)^{\mathsf{T}},\]
\[\mathbf{D}_S=(D_J,D_{JK},D_K,d_1,d_2)^{\mathsf{T}},\]
the reference transform is
\[\mathbf{D}_{R'}=\mathbf{Q}_{R'}^{-1}\mathbf{Q}_R\mathbf{D}_R,\]
where  $\mathbf{Q}_R=\mathbf{Q}_A(R)$  for A reduction and  $\mathbf{Q}_R=\mathbf{Q}_S(R)$  for S reduction.

For S reduction, the explicit implemented matrices are derived from S-Hamiltonian coefficient matching
\[\mathbf{T}_1=\frac{1}{2}\left[\mathrm{coeff}(J_a^2J_b^2)+\mathrm{coeff}(J_a^2J_c^2)+\mathrm{coeff}(J_b^2J_c^2)\right]\]

The quartic tensor algorithm is implemented as
\[\mathbf{D} \rightarrow \mathbf{M}_4 \mathbf{D} \rightarrow \mathbf{P} \mathbf{D} \rightarrow \mathbf{M}_4' \mathbf{D}'\]
with pseudoinverse  $\mathbf{M}_4^+$  and representation permutation matrix  $\mathbf{P}$ .

\section*{Implemented Sextic Tensor Model}
Sextic Watson vector (A notation):
\[\mathbf{H}=(H_J,H_{JK},H_{KJ},H_K,h_1,h_2,h_3)^{\mathsf{T}}.\]
Sextic tensor vector:
\[\boldsymbol{\phi}=(\phi_{aaaaaa},\phi_{bbbbbb},\phi_{cccccc},\phi_{aaaabb},\phi_{aaaacc},\phi_{bbbbaa},\phi_{bbbbaa},\phi_{bbbbaa},\phi_{bbbbaa})^{\mathsf{T}}\]
Mapping and transform:
\[\mathbf{H}=\mathbf{M}_6\boldsymbol{\phi},\]
\[\boldsymbol{\phi}=\mathbf{M}_6^+\mathbf{H},\]
\[\mathbf{H}'=\mathbf{M}_6'\mathbf{P}_6\boldsymbol{\phi}.\]

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